

NUFORC Data Visualization on JavaScript

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Abstract

For the project, the dataset to be used is from an organization called NUFORC (National UFO Reporting Center) [1], which takes in reports made by people around the world and investigates potential explanations of these phenomena. The dataset contains 80,000 reports of UFO sightings from 1905 to 2014. The dataset is available on HuggingFace [2]. The information contained in the data is: date, city, state, country, shape, duration, and comments. The goal of the project is to interactively visualize data in a way that allows us to gain insights into the distribution of sightings across the world and over time, while also making sure the graphs are easy to understand and interact with. The tools to be used for the project are JavaScript libraries such as p5.js [3] and d3.js [4], which are powerful tools for creating interactive data visualizations on the web, also allowing us to create a more immersive experience by using AR (Augmented Reality) as another visualization method.

Keywords

NUFORC — p5.js — d3.js

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1. Design UX

The idea for this data visualization is an immersive experience where the user can feel like they are in the middle of the data, being able to interact with it and explore it in a more intuitive way. For that reason, the use of AR is a key aspect of the design, allowing the user to interact with the data in a more natural way or in intergalactic scenarios.

1. Background of the page is dark, with a touch of blue, to create a mysterious and immersive atmosphere.
2. The font is in a light color, maybe a bit glowing, to contrast with the dark background and make it easier to read.
3. The graphs are big and interactive, allowing the user to explore the data in a more intuitive way. The use of AR allows the user to interact with the data in a more natural way or in intergalactic scenarios.

Important things to consider are the accessibility of the graphs, making sure to take into account color blindness and other visual impairments, as well as making sure the graphs are easy to understand and interact with. This includes the use of clear labels, tooltips, and other interactive elements that can help the user understand the data better. For the design of the graphs, a consistent color scheme will be used, avoiding colors such as red and green together, or yellow for text, to make sure the elements are easy to distinguish and read. Text should also be big enough to be read easily.

2. Visualization Ideas

2.1 Calendar Heatmap

One of the methods to analyse the data could be a calendar heatmap, to view the distribution of sightings over the years. Information to be gathered could be specific increases in sightings, correlating with historical events, or specific seasons of the year where sightings are more common. The calendar heatmap could be filtered by year.

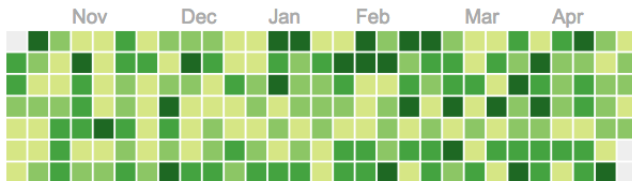


Figure 1. Example of a calendar heatmap [5]

2.2 Geographical Heatmap

In order to view the distribution of sightings across the world, a geographical heatmap could be used. This would allow us to see if there are specific areas where sightings are more common, and which countries are the main contributors. Here is where AR could be useful to visualize the data in a more interactive way, being able to view specific angles of the heatmap by interacting with the camera.

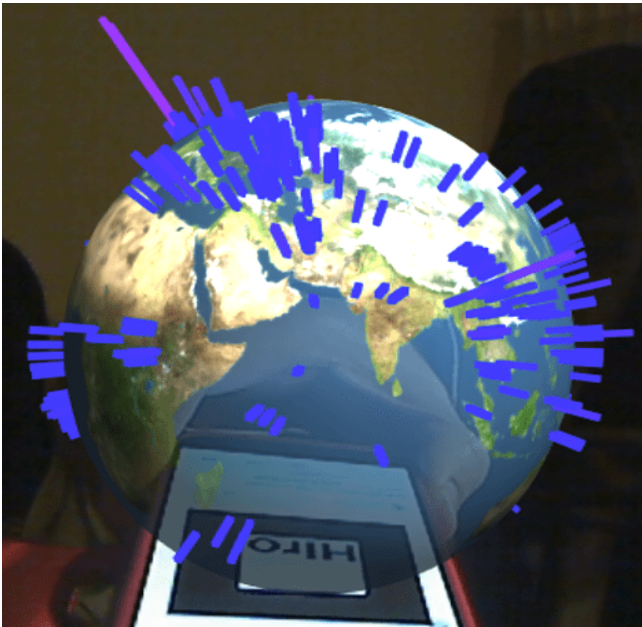


Figure 2. Example of a geographical heatmap [6]

2.3 Bubble Chart

A bubble chart could be used to visualize the most common words used in the comments section of the reports. This could give us insight into the common themes and descriptions used by witnesses. Potentially being able to filter the bubble chart

by features such as shape, duration, or location could provide more specific insights into the data.

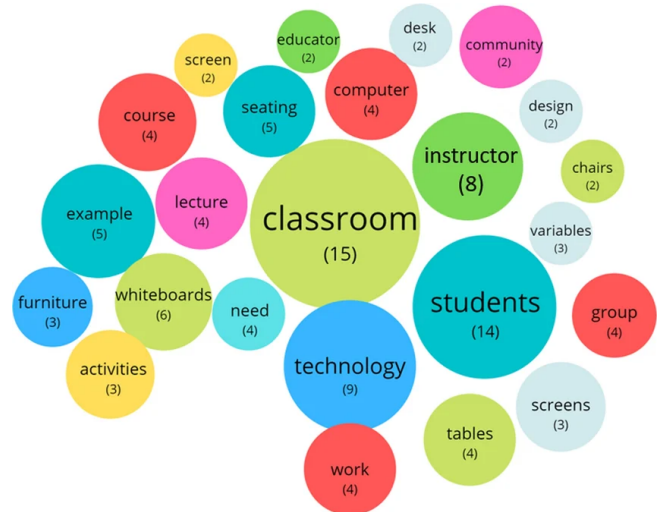


Figure 3. Example of a bubble chart[7]

References

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